

Healthcare Operational Reasoning Benchmark Suite

Version 1.0 Portfolio Overview

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An original benchmark suite for evaluating AI operational reasoning across realistic healthcare workflows.

Can an AI distinguish apparent operational readiness from actual executable readiness?

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Contents

Executive Summary
Suite at a Glance
Motivation
Why Nursing Matters
Design Philosophy
Methodology
Benchmark Index
Benchmark Summaries
Common Failure Modes
Lessons Learned
What Changed My Thinking
Scope and Limitations
Future Directions
About Meredith Reese, RN
Version Information

Executive Summary

Healthcare AI systems increasingly participate in workflows where safe decision-making depends on coordinating multiple evolving information sources rather than retrieving isolated facts.

The Healthcare Operational Reasoning Benchmark Suite is an original benchmark suite designed to evaluate operational reasoning across realistic healthcare workflows. The suite emphasizes prioritization, ownership boundaries, workflow dependencies, temporal reasoning, source provenance, and resource constraints while requiring AI systems to distinguish apparent operational readiness from actual executable readiness.

Rather than evaluating medical knowledge alone, the suite focuses on operational judgment in complex healthcare environments. Each benchmark simulates a short operational decision window and includes fictional source documents, system instructions, a task prompt, and strict grading guidelines.

The suite asks one central question:

Can an AI distinguish apparent operational readiness from actual executable readiness while making operationally sound decisions under realistic healthcare constraints?

Suite at a Glance

12	336	12
Benchmark workflows	Fictional operational source documents	Healthcare operational domains
8	30-45 min	0
Core capability categories	Estimated human evaluation time per benchmark	Real patients, PHI, or confidential clinical records

Healthcare operations reward appropriate restraint as much as decisive action.

Core Capabilities Evaluated

Multi-document reasoning	Operational prioritization
Temporal reasoning	Workflow coordination
Resource allocation	Ownership boundaries
Source provenance	Uncertainty management

Motivation

This project began with a simple question:

What kinds of healthcare reasoning consistently challenge modern AI systems?

Initial exploration focused on documentation review and chart-based reasoning. While many of those tasks produced strong model performance, operational planning tasks involving multiple staff members, competing priorities, evolving workflows, ownership boundaries, and incomplete information consistently exposed richer reasoning failures.

Those observations became the foundation for this benchmark suite.

The project evolved from testing whether a model could retrieve the correct fact to evaluating whether it could make the best operational decision when several actions appeared plausible but only some were realistically executable.

Why Nursing Matters

This benchmark suite was shaped by nursing experience.

Rather than emphasizing isolated factual questions, many scenarios reflect the cognitive demands of real healthcare operations:

- competing priorities
- incomplete information
- evolving workflow
- multiple operational stakeholders
- realistic resource constraints
- continual reprioritization

Many design decisions were influenced by the style of prioritization familiar from nursing education and bedside clinical practice, where several actions may be reasonable but one represents the best operational decision given the circumstances.

In that sense, the benchmarks evaluate more than whether AI knows healthcare terminology. They evaluate whether AI can navigate the cognitive load of healthcare operations.

Design Philosophy

Healthcare operations reward appropriate restraint as much as decisive action.

Operational success depends not only on identifying possible actions, but on determining which actions are realistically executable within existing workflow constraints.

Healthcare workflows often contain artifacts that appear to indicate readiness: green dashboards, scheduled appointments, clean rooms, completed checklists, signed orders, available beds, and copied-forward documentation. In practice, those indicators are often insufficient. Execution may still depend on owner-confirmed decisions, staffing, transport, equipment, safety gates, receiving-unit acceptance, medication readiness, or updated clinical status.

This benchmark suite evaluates whether AI systems can recognize that difference.

The suite intentionally emphasizes realistic ambiguity over artificial complexity. A high-quality response is not the one that sounds the most complete; it is the one that makes only the operational commitments supported by the available evidence.

Methodology

Healthcare Operational Reasoning Benchmark Methodology

Healthcare operations require decisions that extend beyond factual knowledge or document summarization. Coordinators must integrate evolving information from multiple operational sources while accounting for workflow constraints, ownership boundaries, limited resources, and patient safety.

This benchmark suite evaluates whether AI systems produce operationally realistic recommendations under those conditions.

Core Evaluation Principle

Operational readiness cannot be inferred from apparent readiness.

Healthcare environments contain many indicators that appear to suggest readiness:

- Green dashboards
- Scheduled appointments
- Completed checklists
- Ready rooms
- Copied-forward documentation
- Pending discharges
- Available beds
- Completed orders

These indicators often appear sufficient when viewed individually. In practice, operational execution frequently depends on additional owner-confirmed decisions, resource availability, evolving workflow constraints, or unresolved dependencies documented elsewhere.

The benchmarks evaluate whether an AI system recognizes that distinction.

Benchmark Construction Process

1. Select an operational workflow

Each benchmark begins with a realistic healthcare workflow, such as emergency department triage, sterile processing, home health routing, hospice on-call routing, medication reconciliation, case management, or primary care triage.

2. Create a fictional operational environment

Each benchmark includes fictional but operationally plausible documents representing the types of information available to healthcare coordinators. These may include operational boards, staffing assignments, secure messages, callback logs, nursing notes, scheduling views, transport updates, equipment status, pharmacy updates, and huddle agendas.

3. Introduce realistic operational uncertainty

Benchmarks intentionally include evolving operational conditions such as conflicting documentation, copied-forward information, pending owner decisions, staffing changes, workflow bottlenecks, and unresolved dependencies.

4. Require prioritization among plausible actions

The strongest scenarios are not built around a single obvious correct answer. They require the model to weigh multiple operationally plausible actions and identify what can realistically happen next.

5. Evaluate operational reasoning

Models are evaluated on whether they distinguish confirmed information from apparent readiness, respect ownership boundaries, identify unresolved dependencies, prioritize realistically, allocate limited resources appropriately, and avoid unsupported operational commitments.

Evaluation Philosophy

Healthcare operations require balancing decisiveness with appropriate restraint. A high-quality operational plan does not simply recommend the next possible action. It identifies the next action that is realistically executable based on available evidence while explicitly recognizing unresolved dependencies that remain outside the coordinator's authority.

For this reason, the benchmark suite rewards operational judgment rather than confident completion.

Benchmark Index

The v1.0 suite contains twelve benchmark workflows. Each benchmark includes a use case/system prompt, fictional operational source documents, strict grading guidelines, and a benchmark design brief.

ID	Benchmark	Clinical Domain	Operational Window	Source Docs
HORB-001	Labor & Delivery Charge Nurse / OB Triage Flow	Labor & Delivery Operations	5:30 PM to 5:45 PM	28
HORB-002	Sterile Processing / OR Case-Cart Readiness Flow	Sterile Processing and Perioperative Operations	1:10 PM to 1:15 PM	28
HORB-003	Home Health Visit Coordination / Field Nursing Route	Home Health Operations	12:30 PM to 12:45 PM	28
HORB-004	Outpatient Infusion Center / Oncology Treatment Readiness	Outpatient Infusion and Oncology Operations	8:30 AM to 8:45 AM	28
HORB-005	Hospice On-Call / Symptom Crisis & Visit Routing	Hospice Operations	4:10 PM to 4:15 PM	28
HORB-006	Emergency Department Triage / Room Assignment & Ambulance Offload Flow	Emergency Department Operations	6:12 PM to 6:25 PM	28
HORB-007	Perioperative OR Add-On / Turnover Flow	Perioperative Operations	2:14 PM to 2:20 PM	28
HORB-008	Behavioral Health Crisis Unit / Psychiatric Intake & Safety Placement	Behavioral Health Operations	7:14 PM to 7:30 PM	28
HORB-009	Hospital Pharmacy / Medication Reconciliation & Discharge Readiness	Hospital Pharmacy and Transitions of Care	9:30 AM to 9:45 AM	28
HORB-010	Hospital Case Management / Discharge Disposition & Throughput	Case Management and Discharge Planning	11:30 AM to 11:45 AM	28
HORB-011	Perioperative Pre-Admission Testing / Day-of-Surgery Clearance	Perioperative Pre-Admission Testing	3:10 PM to 3:20 PM	28
HORB-012	Primary Care Same-Day Triage / Schedule Routing Flow	Primary Care Operations	1:18 PM to 1:30 PM	28

Benchmark Summaries

Each benchmark summary below is a portfolio-facing overview. The complete benchmark files are included in the v1.0 project package.

HORB-001 | Labor & Delivery Charge Nurse / OB Triage Flow

Evaluating operational reasoning under uncertainty in labor & delivery operations.

Clinical Domain	Labor & Delivery Operations
Operational Setting	Hospital-based Labor & Delivery unit
Operational Window	5:30 PM to 5:45 PM before the Labor and Delivery flow huddle.
Primary Capability	Operational reasoning under uncertainty in a multidisciplinary obstetric workflow.
Materials	Use case/system prompt, grading guidelines, and 28 fictional operational source documents

Purpose

Evaluates whether an AI system can coordinate a Labor & Delivery flow window involving OB triage arrivals, labor-room admissions, postpartum transfers, scheduled inductions, possible urgent C-section evaluation, fetal monitoring changes, anesthesia coverage, Mother-Baby capacity, transport limits, medication readiness, and OR readiness.

Can the model distinguish apparent readiness from executable readiness while producing an operationally realistic plan for this workflow?		
Operational Actors	Reasoning Challenges	Typical Failure Modes
- Charge nurse - OB providers - Labor nurses - Anesthesia - OB OR team - Mother-Baby receiving unit - Transport - Pharmacy / blood bank	- Competing maternal/fetal priorities - Labor-room and postpartum bed availability - Fetal monitoring changes - Anesthesia and OR readiness - GBS/medication readiness - Transport limits - Receiving-unit acceptance - Interdisciplinary communication	- Premature labor-room assignment - Assuming a clean room or open OB OR is usable - Ignoring fetal-monitoring or anesthesia dependencies - Overlooking postpartum receiving constraints - Unrealistic sequencing before the huddle - Unsupported assumptions from green board status

Design Notes

This benchmark was designed to evaluate operational judgment rather than factual recall. It presents multiple operationally plausible actions and requires prioritization based on safety, workflow constraints, source provenance, and confirmed operational readiness. The benchmark fits the suite's core design principle: operational readiness cannot be inferred from apparent readiness.

Difficulty: Expert | Estimated evaluation time: Approximately 30–45 minutes.

HORB-002 | Sterile Processing / OR Case-Cart Readiness Flow

Evaluating operational reasoning under uncertainty in sterile processing and perioperative operations.

Clinical Domain	Sterile Processing and Perioperative Operations
Operational Setting	Hospital sterile processing department and OR case-cart release window
Operational Window	1:10 PM to 1:15 PM before the SPD/OR readiness huddle.
Primary Capability	Operational readiness reasoning for sterile processing and OR case-cart release.
Materials	Use case/system prompt, grading guidelines, and 28 fictional operational source documents

Purpose

Evaluates whether an AI system can reason through case-cart release decisions involving sterilizer cycle status, biological indicators, loaner/vendor documentation, preference-card changes, instrument counts, scope reprocessing, missing instruments, partial carts, runner availability, and OR communication.

Can the model distinguish apparent readiness from executable readiness while producing an operationally realistic plan for this workflow?		
Operational Actors	Reasoning Challenges	Typical Failure Modes
- SPD lead - OR liaison - OR charge / room staff - Sterile processing staff - Vendor / loaner representative - Runner / transport - Quality release owner - Surgeon or service representative	- Sterilizer cycle and biological indicator release - Loaner and vendor documentation - Preference-card changes - Instrument count discrepancies - Scope reprocessing status - Runner availability - OR acceptance - Room pressure versus release safety	- Releasing a cart from a green row alone - Treating vendor verbal confirmation as full readiness - Ignoring biological indicator or count reconciliation - Assuming room-ready OR pressure overrides release gates - Overlooking runner availability - Unsupported release before OR acceptance

Design Notes

This benchmark was designed to evaluate operational judgment rather than factual recall. It presents multiple operationally plausible actions and requires prioritization based on safety, workflow constraints, source provenance, and confirmed operational readiness. The benchmark fits the suite's core design principle: operational readiness cannot be inferred from apparent readiness.

Difficulty: Expert | Estimated evaluation time: Approximately 30–45 minutes.

HORB-003 | Home Health Visit Coordination / Field Nursing Route

Evaluating operational reasoning under uncertainty in home health operations.

Clinical Domain	Home Health Operations
Operational Setting	Home health agency field-visit routing and coordination window
Operational Window	12:30 PM to 12:45 PM before the Home health route huddle.
Primary Capability	Field-route operational reasoning under uncertainty.
Materials	Use case/system prompt, grading guidelines, and 28 fictional operational source documents

Purpose

Evaluates whether an AI system can coordinate home health visit routing while balancing start-of-care readiness, wound VAC supply and order issues, urgent medication-safety messages, field nurse and LPN scope, access problems, caregiver availability, DME delivery, time-sensitive labs, route geography, and limited office-call capacity.

Can the model distinguish apparent readiness from executable readiness while producing an operationally realistic plan for this workflow?		
Operational Actors	Reasoning Challenges	Typical Failure Modes
- Clinical coordinator - Field RN - LPN - Scheduler - Supervisor - Patient / caregiver - DME provider - Hospital discharge team	- Start-of-care readiness - Wound VAC supply and order status - Medication safety concerns - Clinician scope limits - Patient access and caregiver availability - DME/oxygen delivery - Route geography - Limited office-call capacity	- Dispatching a nurse before access or orders are confirmed - Assigning work outside clinician scope - Treating scheduled visit as visit-ready - Ignoring DME or supply dependencies - Prioritizing geography over clinical/operational risk - Assuming caregiver availability from old notes

Design Notes

This benchmark was designed to evaluate operational judgment rather than factual recall. It presents multiple operationally plausible actions and requires prioritization based on safety, workflow constraints, source provenance, and confirmed operational readiness. The benchmark fits the suite's core design principle: operational readiness cannot be inferred from apparent readiness.

Difficulty: Expert | Estimated evaluation time: Approximately 30–45 minutes.

HORB-004 | Outpatient Infusion Center / Oncology

Treatment Readiness

Evaluating operational reasoning under uncertainty in outpatient infusion and oncology operations.

Clinical Domain	Outpatient Infusion and Oncology Operations
Operational Setting	Ambulatory infusion center treatment-readiness window
Operational Window	8:30 AM to 8:45 AM before the Infusion treatment-readiness huddle.
Primary Capability	Treatment-readiness reasoning across pharmacy, nursing, authorization, and chair capacity constraints.
Materials	Use case/system prompt, grading guidelines, and 28 fictional operational source documents

Purpose

Evaluates whether an AI system can coordinate an outpatient infusion readiness window involving chemotherapy and biologic treatment readiness, lab parameters, pharmacy compounding, prior authorization, blood product compatibility, port access, chair capacity, nurse assignments, and patient communication.

Can the model distinguish apparent readiness from executable readiness while producing an operationally realistic plan for this workflow?		
Operational Actors	Reasoning Challenges	Typical Failure Modes
• Infusion charge nurse • Infusion RN • Oncology provider • Pharmacy / compounding • Prior authorization team • Lab / blood bank • Patient access / front desk • Patient / caregiver	• Lab parameter verification • Pharmacy compounding readiness • Prior authorization status • Chair and nurse capacity • Blood product compatibility • Port access or access device readiness • Premedication requirements • Patient communication and arrival timing	• Treating chair availability as treatment readiness • Assuming pharmacy can compound without verified release • Ignoring prior authorization or lab dependencies • Starting treatment based on schedule status alone • Overlooking blood product or access issues • Unrealistic chair sequencing

Design Notes

This benchmark was designed to evaluate operational judgment rather than factual recall. It presents multiple operationally plausible actions and requires prioritization based on safety, workflow constraints, source provenance, and confirmed operational readiness. The benchmark fits the suite's core design principle: operational readiness cannot be inferred from apparent readiness.

Difficulty: Expert | Estimated evaluation time: Approximately 30–45 minutes.

HORB-005 | Hospice On-Call / Symptom Crisis & Visit Routing

Evaluating operational reasoning under uncertainty in hospice operations.

Clinical Domain	Hospice Operations
Operational Setting	Hospice on-call symptom crisis and visit-routing window
Operational Window	4:10 PM to 4:15 PM before the Hospice on-call route huddle.
Primary Capability	Hospice visit-routing and symptom-crisis operational reasoning.
Materials	Use case/system prompt, grading guidelines, and 28 fictional operational source documents

Purpose

Evaluates whether an AI system can coordinate hospice on-call routing while balancing symptom crisis calls, comfort-kit readiness, medication order ownership, DME/oxygen status, caregiver availability, nurse visit capacity, after-hours escalation, and realistic visit routing.

Can the model distinguish apparent readiness from executable readiness while producing an operationally realistic plan for this workflow?		
Operational Actors	Reasoning Challenges	Typical Failure Modes
- Hospice on-call coordinator - Hospice RN - Hospice physician / NP - Pharmacy - DME / oxygen provider - Patient / caregiver - Social work / chaplain as applicable - Triage or after-hours team	- Symptom crisis prioritization - Comfort-kit versus active medication orders - DME and oxygen functionality - Caregiver access and safety - Visit routing capacity - After-hours provider ownership - Urgent versus routine needs - Unresolved admission or visit dependencies	- Routing a visit before medication or DME gates are clear - Treating comfort-kit presence as medication readiness - Ignoring caregiver access or safety issues - Overcommitting limited hospice RN capacity - Assuming scheduled admission equals admission-ready - Under-escalating symptom crisis calls

Design Notes

This benchmark was designed to evaluate operational judgment rather than factual recall. It presents multiple operationally plausible actions and requires prioritization based on safety, workflow constraints, source provenance, and confirmed operational readiness. The benchmark fits the suite's core design principle: operational readiness cannot be inferred from apparent readiness.

Difficulty: Expert | Estimated evaluation time: Approximately 30–45 minutes.

HORB-006 | Emergency Department Triage / Room Assignment & Ambulance Offload Flow

Evaluating operational reasoning under uncertainty in emergency department operations.

Clinical Domain	Emergency Department Operations
Operational Setting	Emergency Department triage, rooming, and EMS offload window
Operational Window	6:12 PM to 6:25 PM before the ED triage and rooming huddle.
Primary Capability	Emergency Department operational prioritization and room-assignment reasoning.
Materials	Use case/system prompt, grading guidelines, and 28 fictional operational source documents

Purpose

Evaluates whether an AI system can coordinate ED triage and room assignment while balancing patient acuity, room suitability, ambulance offload, nurse/tech coverage, safety risks, isolation, monitoring needs, and evolving waiting-room information.

Can the model distinguish apparent readiness from executable readiness while producing an operationally realistic plan for this workflow?		
Operational Actors	Reasoning Challenges	Typical Failure Modes
• Triage nurse • Charge nurse • ED physicians / APPs • Primary nurses • Techs • EMS • Registration • Security or sitter coverage as needed	• Competing ED acuity • Room suitability and monitoring needs • Ambulance offload pressure • Staffing and sitter limitations • Isolation/safety requirements • Waiting-room status changes • Patient deterioration signals • Multiple plausible rooming choices	• Rooming a patient based only on green status • Ignoring monitoring or safety needs • Treating open room as usable without staffing • Premature EMS offload assignment • Unrealistic simultaneous room moves • Under-prioritizing new high-risk symptoms

Design Notes

This benchmark was designed to evaluate operational judgment rather than factual recall. It presents multiple operationally plausible actions and requires prioritization based on safety, workflow constraints, source provenance, and confirmed operational readiness. The benchmark fits the suite's core design principle: operational readiness cannot be inferred from apparent readiness.

Difficulty: Expert | Estimated evaluation time: Approximately 30–45 minutes.

HORB-007 | Perioperative OR Add-On / Turnover Flow

Evaluating operational reasoning under uncertainty in perioperative operations.

Clinical Domain	Perioperative Operations
Operational Setting	Afternoon OR add-on and turnover coordination window
Operational Window	2:14 PM to 2:20 PM before the Perioperative add-on flow huddle.
Primary Capability	OR add-on sequencing and turnover operational reasoning.
Materials	Use case/system prompt, grading guidelines, and 28 fictional operational source documents

Purpose

Evaluates whether an AI system can coordinate perioperative add-on and turnover flow involving OR room status, case urgency, anesthesia coverage, PACU capacity, instrumentation, patient readiness, vendor/equipment needs, transport, and short-window huddle preparation.

Can the model distinguish apparent readiness from executable readiness while producing an operationally realistic plan for this workflow?		
Operational Actors	Reasoning Challenges	Typical Failure Modes
- OR charge nurse - Anesthesia lead - Surgeons / services - PACU charge - Pre-op nurses - SPD / sterile processing - Transport - Vendor / equipment support	- Add-on urgency versus readiness - Turnover status and room release - Anesthesia and PACU availability - Instrumentation and equipment readiness - Patient transport and consent readiness - Vendor dependencies - Competing room pressure - Short huddle window	- Choosing a first add-on from board status alone - Assuming turnover means case-ready - Ignoring PACU or anesthesia constraints - Opening instruments prematurely - Sequencing cases without owner release - Overcommitting transport or room resources

Design Notes

This benchmark was designed to evaluate operational judgment rather than factual recall. It presents multiple operationally plausible actions and requires prioritization based on safety, workflow constraints, source provenance, and confirmed operational readiness. The benchmark fits the suite's core design principle: operational readiness cannot be inferred from apparent readiness.

Difficulty: Expert | Estimated evaluation time: Approximately 30–45 minutes.

HORB-008 | Behavioral Health Crisis Unit / Psychiatric Intake & Safety Placement

Evaluating operational reasoning under uncertainty in behavioral health operations.

Clinical Domain	Behavioral Health Operations
Operational Setting	Behavioral health crisis unit intake and placement window
Operational Window	7:14 PM to 7:30 PM before the Behavioral health placement huddle.
Primary Capability	Psychiatric intake placement reasoning with safety, legal, and staffing constraints.
Materials	Use case/system prompt, grading guidelines, and 28 fictional operational source documents

Purpose

Evaluates whether an AI system can coordinate behavioral health intake and placement while balancing legal status, medical clearance, contraband/security search, observation level, sitter coverage, room safety, unit acceptance, transportation, and evolving patient behavior.

Can the model distinguish apparent readiness from executable readiness while producing an operationally realistic plan for this workflow?		
Operational Actors	Reasoning Challenges	Typical Failure Modes
- Behavioral health charge nurse - Intake clinician - Psychiatrist / provider - Security - ED or referring unit - Sitter / observer - Nursing staff - Transport	- Legal and voluntary/involuntary status - Medical clearance - Security and contraband search - Observation level requirements - Room safety and staffing - Evolving behavior and risk - Unit acceptance - Transport timing	- Treating calm behavior as placement-ready - Assuming clean room equals safe placement - Ignoring legal status or medical clearance - Underestimating observation staffing - Premature placement before security search - Overconfidence from green intake board status

Design Notes

This benchmark was designed to evaluate operational judgment rather than factual recall. It presents multiple operationally plausible actions and requires prioritization based on safety, workflow constraints, source provenance, and confirmed operational readiness. The benchmark fits the suite's core design principle: operational readiness cannot be inferred from apparent readiness.

Difficulty: Expert | Estimated evaluation time: Approximately 30–45 minutes.

HORB-009 | Hospital Pharmacy / Medication Reconciliation & Discharge Readiness

Evaluating operational reasoning under uncertainty in hospital pharmacy and transitions of care.

Clinical Domain	Hospital Pharmacy and Transitions of Care
Operational Setting	Medication reconciliation and discharge readiness huddle window
Operational Window	9:30 AM to 9:45 AM before the Medication reconciliation huddle.
Primary Capability	Medication reconciliation and discharge-readiness operational reasoning.
Materials	Use case/system prompt, grading guidelines, and 28 fictional operational source documents

Purpose

Evaluates whether an AI system can coordinate transitions-of-care medication readiness while balancing discharge orders, pharmacist verification, renal dosing, duplicate therapies, anticoagulation, insulin, counseling, interpreter needs, prior authorization, meds-to-beds, and discharge timing.

Can the model distinguish apparent readiness from executable readiness while producing an operationally realistic plan for this workflow?		
Operational Actors	Reasoning Challenges	Typical Failure Modes
- Transitions pharmacist - Pharmacy technician - Discharge nurse - Prescriber / hospitalist - Case manager - Interpreter services - Meds-to-beds courier - Patient / caregiver	- Incomplete medication reconciliation - High-risk medication review - Renal dosing or duplicate therapy - Anticoagulation and insulin safety - Prior authorization status - Counseling and interpreter needs - Meds-to-beds delivery - Discharge timing pressure	- Treating signed discharge order as medication-ready - Equating paid claim with discharge readiness - Ignoring pharmacist verification or counseling needs - Missing high-risk medication conflicts - Overlooking prior authorization or delivery barriers - Releasing discharge before med rec closure

Design Notes

This benchmark was designed to evaluate operational judgment rather than factual recall. It presents multiple operationally plausible actions and requires prioritization based on safety, workflow constraints, source provenance, and confirmed operational readiness. The benchmark fits the suite's core design principle: operational readiness cannot be inferred from apparent readiness.

Difficulty: Expert | **Estimated evaluation time:** Approximately 30–45 minutes.

HORB-010 | Hospital Case Management / Discharge Disposition & Throughput

Evaluating operational reasoning under uncertainty in case management and discharge planning.

Clinical Domain	Case Management and Discharge Planning
Operational Setting	Late-morning discharge disposition and throughput window
Operational Window	11:30 AM to 11:45 AM before the Case management discharge flow huddle.
Primary Capability	Discharge disposition and throughput operational reasoning.
Materials	Use case/system prompt, grading guidelines, and 28 fictional operational source documents

Purpose

Evaluates whether an AI system can coordinate case management discharge planning while balancing discharge orders, SNF or facility acceptance, payer authorization, PASRR/level-of-care requirements, DME, home oxygen, dialysis, wound VAC, caregiver readiness, transport, discharge lounge capacity, and family communication.

Can the model distinguish apparent readiness from executable readiness while producing an operationally realistic plan for this workflow?		
Operational Actors	Reasoning Challenges	Typical Failure Modes
- Case manager - Social worker - Bed control - Discharge nurse - Provider - SNF/facility liaison - Payer / authorization team - DME / oxygen vendor	- Destination acceptance - Payer and level-of-care approvals - DME / oxygen readiness - Dialysis or wound VAC dependencies - Caregiver and family pickup readiness - Transport confirmation - Discharge lounge capacity - Bed throughput pressure	- Counting signed discharge order as executable discharge - Treating facility bed offer as final acceptance - Ignoring payer/PASRR requirements - Assuming DME portal status means home setup ready - Overcommitting transport or discharge lounge capacity - Releasing discharge before receiving handoff

Design Notes

This benchmark was designed to evaluate operational judgment rather than factual recall. It presents multiple operationally plausible actions and requires prioritization based on safety, workflow constraints, source provenance, and confirmed operational readiness. The benchmark fits the suite's core design principle: operational readiness cannot be inferred from apparent readiness.

Difficulty: Expert | Estimated evaluation time: Approximately 30–45 minutes.

HORB-011 | Perioperative Pre-Admission Testing / Day-of-Surgery Clearance

Evaluating operational reasoning under uncertainty in perioperative pre-admission testing.

Clinical Domain	Perioperative Pre-Admission Testing
Operational Setting	Late-afternoon next-day surgery readiness and cancellation-prevention window
Operational Window	3:10 PM to 3:20 PM before the PAT / day-of-surgery readiness huddle.
Primary Capability	Surgical readiness and cancellation-prevention operational reasoning.
Materials	Use case/system prompt, grading guidelines, and 28 fictional operational source documents

Purpose

Evaluates whether an AI system can coordinate pre-admission testing and next-day surgical readiness while balancing anesthesia clearance, anticoagulant holds, GLP-1/NPO updates, respiratory symptoms, abnormal labs, cardiac clearance, interpreter/consent needs, OSA/CPAP recovery planning, vendor/implant readiness, callbacks, and OR schedule preservation.

Can the model distinguish apparent readiness from executable readiness while producing an operationally realistic plan for this workflow?		
Operational Actors	Reasoning Challenges	Typical Failure Modes
- PAT lead - Anesthesia reviewer - Surgeon office - Cardiology / clearance owner - Interpreter services - Vendor / implant representative - OR scheduling - Patient / caregiver	- Anesthesia acceptance - Medication-hold review - NPO/GLP-1 status - New symptoms or abnormal labs - Cardiac clearance - Consent and interpreter readiness - Vendor/implant readiness - Patient callback completion	- Treating PAT complete as surgery cleared - Assuming signed consent means anesthesia acceptance - Ignoring medication-hold or lab gates - Accepting vendor verbal workaround as final readiness - Preserving schedule without clearance support - Overlooking patient callback needs

Design Notes

This benchmark was designed to evaluate operational judgment rather than factual recall. It presents multiple operationally plausible actions and requires prioritization based on safety, workflow constraints, source provenance, and confirmed operational readiness. The benchmark fits the suite's core design principle: operational readiness cannot be inferred from apparent readiness.

Difficulty: Expert | Estimated evaluation time: Approximately 30–45 minutes.

HORB-012 | Primary Care Same-Day Triage / Schedule Routing Flow

Evaluating operational reasoning under uncertainty in primary care operations.

Clinical Domain	Primary Care Operations
Operational Setting	Primary care same-day access and triage-routing window
Operational Window	1:18 PM to 1:30 PM before the Primary care same-day access huddle.
Primary Capability	Ambulatory triage and schedule-routing operational reasoning.
Materials	Use case/system prompt, grading guidelines, and 28 fictional operational source documents

Purpose

Evaluates whether an AI system can coordinate primary care same-day triage and routing while balancing appointment requests, nurse-visit slots, telehealth holds, parking-lot symptoms, portal updates, voicemail symptoms, pediatric calls, allergic-reaction screening, interpreter needs, provider slot capacity, front desk routing, and emergency escalation decisions.

Can the model distinguish apparent readiness from executable readiness while producing an operationally realistic plan for this workflow?		
Operational Actors	Reasoning Challenges	Typical Failure Modes
• Triage nurse • Provider • Front desk / scheduling • Call center • Interpreter services • Patient / caregiver • Pediatric or specialty support as relevant • Clinic leadership / huddle team	• Red-flag symptom review • Same-day versus ED escalation • Nurse-scope fit • Telehealth appropriateness • Parking-lot/waiting-room concerns • Interpreter or caregiver needs • Provider slot capacity • Multiple patient requests competing for access	• Treating appointment slot as destination safety • Routing to nurse visit without scope fit • Ignoring red-flag symptoms • Assuming patient-requested visit type is appropriate • Using portal response when direct escalation is needed • Overcommitting limited same-day slots

Design Notes

This benchmark was designed to evaluate operational judgment rather than factual recall. It presents multiple operationally plausible actions and requires prioritization based on safety, workflow constraints, source provenance, and confirmed operational readiness. The benchmark fits the suite's core design principle: operational readiness cannot be inferred from apparent readiness.

Difficulty: Expert | Estimated evaluation time: Approximately 30–45 minutes.

Common Failure Modes

Across the benchmark suite, recurring model failure patterns include the following.

Apparent Readiness Bias

Treating green board status, scheduled slots, clean rooms, completed checklists, or visible requests as proof that an operational action is ready to execute.

Premature Operational Commitment

Selecting a first move, first discharge, first placement, first route, or first release before owner-confirmed gates are closed.

Ownership Boundary Violations

Making decisions that belong to another operational owner, such as anesthesia, pharmacy, provider, receiving unit, sterile processing release, payer authorization, or legal-status confirmation.

Workflow Compression

Compressing unrealistic amounts of communication, verification, routing, and execution into a short operational window.

Source Provenance Failure

Over-trusting copied-forward documentation, clerical summaries, dashboard rows, or schedule-derived status over newer owner-confirmed information.

Resource Blindness

Ignoring limited transport, staffing, room, chair, equipment, pharmacy, interpreter, or observation resources.

False Certainty Under Uncertainty

Resolving ambiguity into a confident plan rather than identifying what remains unresolved and who owns the next step.

Lessons Learned

Building this suite led to several observations about healthcare AI evaluation and benchmark design.

Operational reasoning differs from documentation review

Documentation review and chart-based reasoning often produced stronger model performance than expected. Operational planning under uncertainty produced richer evaluation behavior because the model had to coordinate evolving information, people, resources, and workflow constraints.

Realistic workflow constraints matter

The benchmarks became more discriminative when they included staff roles, ownership boundaries, resource limitations, and time constraints. These elements shifted the task from information retrieval to operational judgment.

Prompt simplification increased difficulty

Early iterations used more detailed prompts and system instructions. Simplifying the task framing often increased benchmark strength because the model had to derive the relevant operational reasoning from the source documents rather than relying on explicit prompt guidance.

Multiple plausible actions produced richer evaluation

Scenarios with one clearly correct operational choice were often too easy. The strongest benchmarks required prioritizing among multiple reasonable actions, similar to nursing-style prioritization where several choices may be defensible but one is most appropriate given the situation.

Operational restraint is a key evaluation signal

A strong response often involved identifying what should not be done yet. The suite rewards appropriate uncertainty, escalation, and handoff rather than unsupported confidence.

What Changed My Thinking

Initially, I assumed that increasing document complexity would create stronger evaluations.

Instead, I found that simplifying task instructions while increasing operational realism often produced richer reasoning behavior.

Scenarios with a single clearly correct operational choice frequently measured information retrieval more than operational judgment. The strongest evaluations emerged when multiple operationally plausible actions existed simultaneously, requiring prioritization rather than simple identification of the correct answer.

This shifted the project from testing whether a model could find the right fact to testing whether it could make the best operational decision under uncertainty.

Scope and Limitations

Purpose

The Healthcare Operational Reasoning Benchmark Suite was developed to evaluate how AI systems perform when operational decisions require integrating multiple evolving information sources under realistic healthcare workflow constraints.

The suite is intended to assess operational reasoning rather than factual recall or medical knowledge alone.

Scope

This benchmark suite evaluates an AI system's ability to:

- Reason across multiple operational documents simultaneously.
- Distinguish apparent readiness from executable operational readiness.
- Respect workflow ownership and decision boundaries.
- Prioritize among competing operational demands.
- Manage uncertainty when information is incomplete or evolving.
- Produce operationally realistic recommendations rather than simply identifying isolated facts.

What This Project Does Not Evaluate

This benchmark suite is not intended to evaluate:

- Diagnostic accuracy.
- Clinical treatment recommendations.
- Medical knowledge recall.
- Guideline memorization.
- Board-style examination performance.
- Clinical competence independent of operational context.

Instead, the suite focuses specifically on operational reasoning within healthcare workflows.

Fictional Content

All benchmark materials are fictional and were created solely for evaluation and portfolio purposes. No real patients, health records, healthcare organizations, or confidential clinical information are included.

Current Limitations

This project represents an independent benchmark design effort and should be understood within that context.

Current limitations include:

- The benchmark suite has not yet undergone formal inter-rater reliability studies.
- Benchmark difficulty has been refined through iterative design rather than large-scale statistical validation.
- Performance analyses across multiple AI models are outside the scope of this portfolio.
- Future revisions may expand domain coverage, improve rubric calibration, and incorporate formal validation methodologies.

Future Directions

Potential future enhancements include:

- Formal benchmark validation with multiple independent evaluators.
- Difficulty calibration across benchmark families.
- Comparative evaluation of multiple AI systems.
- Expanded healthcare operational domains.
- Longitudinal benchmark versioning and maintenance.
- Additional benchmark families focused on other AI capabilities, such as clinical communication, longitudinal chart reasoning, or patient education.

Potential future operational domains include:

- Critical care operations
- Dialysis operations
- Oncology navigation
- Rehabilitation
- Long-term care
- Telemedicine
- Ambulatory surgery
- Population health coordination

About Meredith Reese, RN

I'm Meredith Reese, RN.

This project began as an independent exploration of AI behavior while interacting with large language models during my own time. As the project evolved, I became increasingly interested in how AI systems perform when operational healthcare decisions require prioritization under uncertainty rather than straightforward factual recall.

Drawing on my nursing background, I developed an original benchmark suite centered on operational reasoning across multiple healthcare environments.

Areas of Interest

- Healthcare AI evaluation
- Benchmark design
- Operational reasoning
- Clinical workflow modeling
- Human-AI collaboration
- Model quality assessment

Version Information

Version 1.0

Initial portfolio release of the Healthcare Operational Reasoning Benchmark Suite.

Includes:

- 12 original benchmark domains
- Benchmark design briefs
- Fictional operational source documents
- System instructions and prompts
- Strict grading guidelines
- Methodology documentation
- Failure mode taxonomy
- Lessons learned
- Scope and limitations

Version 1.1 — Planned

Potential improvements:

- Additional rubric refinement
- Expanded benchmark summaries
- Difficulty calibration
- Reviewer-facing scoring sheets
- Enhanced website presentation

Version 2.0 — Future

Potential future directions:

- Cross-model evaluation
- Comparative performance analysis
- New benchmark families
- Formal inter-rater validation
- Expanded healthcare operations domains

Closing Reflection

This project began with curiosity and became a structured portfolio project centered on healthcare operational reasoning. The v1.0 release presents an original suite of fictional healthcare workflow benchmarks designed to test whether AI systems can move beyond document review and produce operationally realistic, appropriately constrained recommendations.

The work was already there. This portfolio teaches it how to introduce itself.